

House Price Prediction Project

1. Project Overview and Objectives

Project Overview

This project builds a machine learning model to predict house prices using features such as area, bedrooms, location, and property type. It includes data exploration, preprocessing, model training, and evaluation using multiple regression algorithms. Different models were compared to identify the most accurate predictor. The analysis also highlights key factors that influence property prices.

Objectives

- To analyze housing data to understand the relationship between property features and house prices.
- To identify key factors such as area, location, number of bedrooms, and property type that influence house prices.
- To build and compare machine learning models to accurately predict house prices.
- To evaluate model performance using metrics such as MAE, RMSE, and R^2 score.
- To generate insights that can help real estate stakeholders make data-driven pricing decisions.

2. Setup and Installation Instructions

Prerequisites

- Python 3.10+
- Jupyter Notebook

Installation Steps

1. Download or clone the project repository from GitHub.
2. Install required libraries:

```
pip install -r requirements.txt
```

3. Place dataset inside /data folder.
4. Launch Jupyter Notebook and open the notebook:
notebooks/house_price_prediction.ipynb

3. Code Structure Explanation

```
HOUSE_PRICE_PREDICTION
|
├── data
|   └── house_prices.csv
|
├── outputs
|   ├── predictions_vs_actual.png
|   └── predictions_vs_actual.csv
|
├── notebook
|   └── house_price_prediction.ipynb
|
├── src
|   ├── data_preprocessing.py
|   ├── model_training.py
|   ├── evaluation.py
|   ├── visualization.py
|   └── linear_regression_scratch.py
|
├── report
|   └── model_evaluation.md
|
├── README.md
├── requirements.txt
└── .gitignore
```

File Descriptions

- data/house_pric.csv
Dataset containing housing features and corresponding house prices.
- outputs/
Folder containing generated visualization such as predictions_vs_actual plot and generated csv file.
- notebooks/house_price_prediction.ipynb
Notebook implementing the full ML workflow: exploration, preprocessing, training, and evaluation.
- src/data_preprocessing.py
Functions for loading data, preparing features, and splitting the dataset.
- src/model_training.py
Training pipelines for Linear Regression, Decision Tree, and Random Forest models.
- src/evaluation.py
Functions for evaluating models using MAE, MSE, RMSE, and R^2 metrics.
- src/visualization.py
Utilities for creating project visualizations.
- src/linear_regression_scratch.py
Custom implementation of Linear Regression.
- report/model_evaluation.md
Report summarizing model results and insights.
- requirements.txt
Specifies all Python dependencies needed to run the project.
- README.md
Project documentation including overview, setup instructions, and usage guide.

4. Screenshots of Working Application

The included screenshots demonstrate:

- The Area vs Price plot shows a strong positive relationship, indicating larger houses have higher prices.
- The Location vs Price plot shows that City Center properties have higher median prices than suburban and rural areas.
- Feature importance analysis identifies Area as the most influential predictor, followed by location features.
- The Actual vs Predicted plot shows predictions closely aligned with actual values, indicating good model accuracy.
- Model evaluation results (**Random Forest R^2 : 0.97**) confirm strong predictive performance.



Fig. 4.1. Relationship between Area and House Price

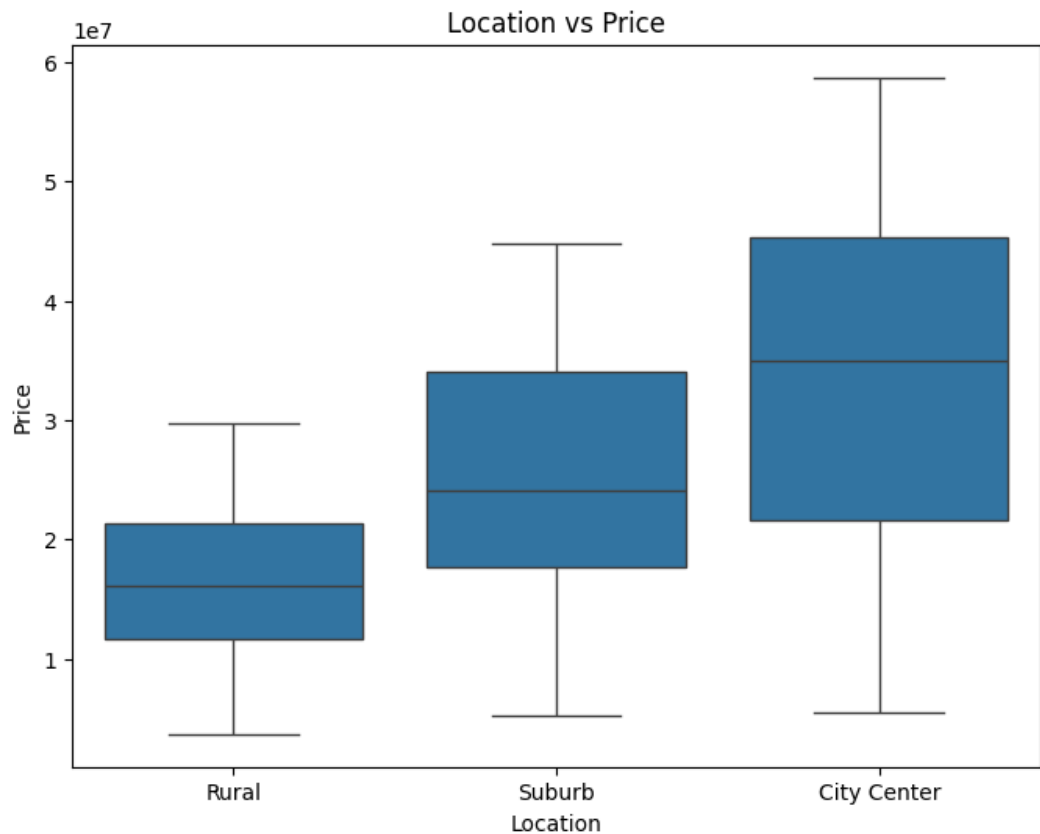


Fig. 4.2. Relationship between House Price and Location

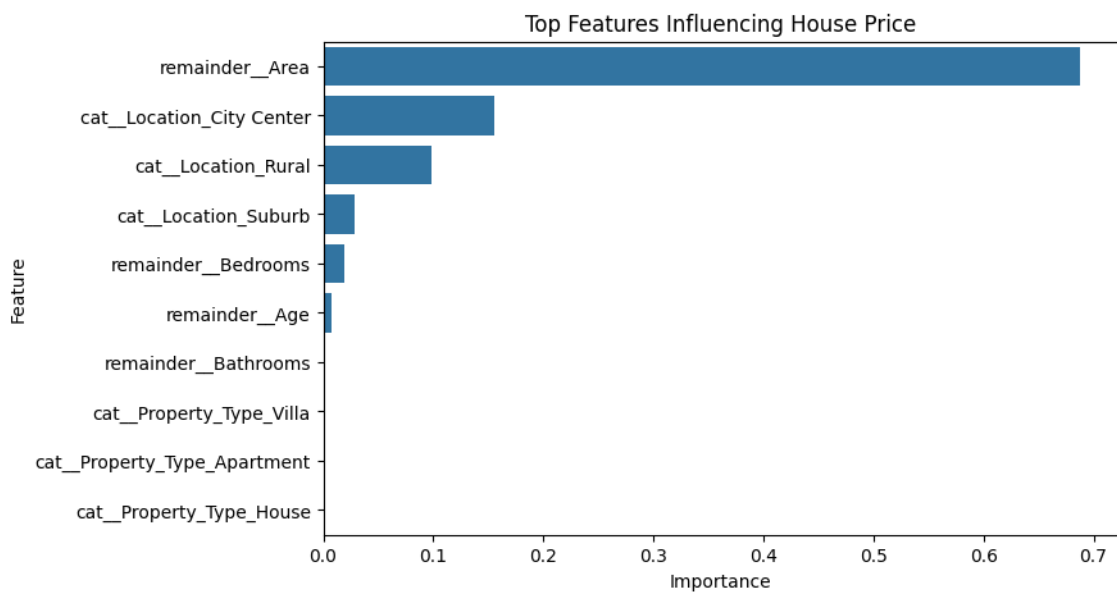


Fig. 4.3. Visualization of Feature Importance



Fig.4.4. Prediction Result

These results validate that the data analysis, machine learning pipeline, and evaluation process were successfully implemented, providing insights into key factors affecting house prices.

5. Technical Requirements Fulfillment Explanation

Category	Technical Requirement	How It Was Implemented
Algorithm	Data loading and preprocessing	Housing dataset loaded using Pandas, checked for missing values, and categorical variables prepared for encoding.
Algorithm	Data cleaning logic	Dataset validated using <code>.info()</code> , <code>.describe()</code> , and <code>.isnull()</code> to ensure correct data types and completeness.

Category	Technical Requirement	How It Was Implemented
Algorithm	Feature engineering	Features such as Area, Bedrooms, Bathrooms, Age, Location, and Property Type were selected for training the prediction model.
Algorithm	Exploratory data analysis	Relationships between features and price analyzed using scatter plots and box plots (Area vs Price, Location vs Price).
Algorithm	Statistical relationship analysis	Feature influence evaluated using Random Forest feature importance to identify the most impactful variables.
Algorithm	Predictive modeling	Linear Regression, Polynomial Regression, Decision Tree, and Random Forest models implemented using Scikit-learn.
Algorithm	Model optimization	Cross-validation and GridSearchCV applied to tune Random Forest hyperparameters and improve model performance.
Data Structure	Tabular data handling	Pandas DataFrames used to store, process, and transform structured housing data.
Data Structure	Feature storage	Encoded categorical features and numerical features converted into arrays for model training.

Category	Technical Requirement	How It Was Implemented
Architecture	Modular project design	Code organized into modular files: data preprocessing, model training, evaluation, visualization, and custom regression implementation.
Architecture	Reproducible ML pipeline	Data preprocessing, feature encoding, model training, and evaluation executed through Scikit-learn pipelines.
Output	Visualization outputs	Visualizations generated including Area vs Price scatter plot, Location vs Price box plot, Feature Importance chart, and Actual vs Predicted plot.
Output	Model evaluation metrics	Models evaluated using MAE, MSE, RMSE, and R^2 Score to compare predictive performance.
Output	Model comparison insights	Random Forest achieved the highest R^2 score (~ 0.97) and lowest prediction error among tested models.
Output	Business insights	Key price drivers identified as Area, Location (City Center), and number of Bedrooms, helping understand real estate pricing patterns.

6. Learning Outcomes

Through this project, the following learning outcomes were achieved:

- Gained practical experience in implementing an end-to-end machine learning workflow using Python.

- Learned data preprocessing techniques including handling categorical variables, feature selection, and train–test splitting.
- Developed skills in exploratory data analysis and data visualization using Matplotlib and Seaborn to understand relationships between housing features and price.
- Implemented and compared multiple machine learning models including Linear Regression, Polynomial Regression, Decision Tree, and Random Forest.
- Applied model evaluation techniques using MAE, MSE, RMSE, and R^2 score to assess predictive performance.
- Understood how feature importance analysis can identify the key factors influencing house price predictions.

7. Conclusion

This project demonstrated how machine learning can be used to predict house prices based on property features. Through data exploration, preprocessing, and model training, key factors influencing house prices were identified. Among the models tested, Random Forest achieved the best performance. The results highlight how data-driven approaches can support better decision-making in the real estate market.